USS Pampanito — AI Docent

Application Overview & Technical Summary

1. What Is This?

SubDocentAI is a voice- and text-enabled Q&A assistant that answers visitor questions about the USS Pampanito (SS-383) and WWII submarine history. It runs as a local HTTPS server on a Mac or small Linux machine, serving a mobile web app to any visitor's phone or a shared kiosk tablet over the local network.

Visitors speak or type a question and receive a spoken, historically accurate answer in seconds, drawn exclusively from verified museum-grade sources. No app download is required. No internet connection is needed during operation.

2. The Problem It Solves

Museum visitors are curious. They stand inside a torpedo room and want to know whether sailors actually slept next to live torpedoes, or how many men were on the boat, or what a depth charge felt like from inside. Fixed audio narration cannot answer these questions; human docents cannot be everywhere at once.

- Curiosity goes unrewarded — visitors leave with unanswered questions.
- Non-English-speaking visitors receive no additional context.
- Staff cannot scale to provide personalised responses to hundreds of visitors daily.

SubDocentAI provides an always-available, on-premises AI guide that responds to natural spoken questions in real time.

3. Visitor Experience — How It Works

Step	What Happens
1. Arrive at a compartment	The visitor enters a compartment (e.g., Control Room, Forward Torpedo Room) and taps to hear the narrated audio for that space.
2. Listen to narration	Pre-recorded audio plays — the same narration currently offered by the Pampanito tour.
3. Ask a question	At any point, the visitor taps a microphone (or record) button and speaks naturally: "How did the torpedo work?" or "Where did the crew sleep?"
4. AI retrieves an answer	The transcript is sent to the on-premises retrieval engine, which searches curated historical sources and assembles a direct spoken answer in under two seconds.
5. Answer plays aloud	The answer is read aloud through the visitor's phone speaker or headphones, with a spoken location prefix when relevant (e.g., "In the Control Room…").

Step	What Happens
6. Follow-ups offered	The guide suggests related questions to keep exploration going.
7. Unanswered questions	If the system cannot find a reliable answer, it says so honestly rather than guessing.

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4. Knowledge Sources

The AI guide answers exclusively from three curated, authoritative corpora. It does not search the internet and cannot fabricate information.

Source	Size	Description
Pampanito Tour Script	213 chunks	The official compartment-by-compartment narration for all 11 interior compartments plus forward and after decks. Each chunk is tagged with a compartment ID, location context, and display citation.
Submarine FAQ Corpus	458 entries	Custom-authored Q&A pairs covering crew life, weapons, engineering, tactics, escape, medical, communications, and Pampanito-specific history. 226 entries (pam_001–pam_226) are Pampanito-focused; 232 entries cover WWII diesel-electric submarines broadly.
DieselSubs Shorts	31 chunks	Concise background explanations on submarine systems, operations, and history, used to supplement answers when the other sources are insufficient.

Total corpus: 702 chunks. Every answer includes a citation identifying which source and chunk the information came from.

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5. Key Features

Feature	Detail
Voice input — iOS/desktop	Browser Web Speech API (`webkitSpeechRecognition`) captures the question locally — no audio upload required.
Voice input — Android	Audio is recorded via MediaRecorder and POSTed to the `/transcribe` endpoint; Groq Whisper (`whisper-large-v3-turbo`) transcribes it server-side (~0.3 s latency). Falls back to a text input row on browsers without microphone support.
Spoken answers	Text-to-speech delivers answers aloud — eyes-free while exploring the boat.
Multilingual audio	Answers available in English, Spanish, French, German, Japanese, and Chinese.

Feature	Detail
Location-aware retrieval	The AI knows which compartment the visitor is standing in and weights results accordingly.
No internet required	Fully self-contained — runs on a Mac or small server on or near the vessel. (Groq Whisper requires internet only if Android transcription is enabled.)
No app download	Progressive Web App served over HTTPS — works in any modern smartphone browser.
Honest fallback	If no reliable answer is found, the system says so rather than guessing. `partial_match` flag signals relevant-but-indirect answers.

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6. Technical Architecture

The application is built on three layers: a browser-based frontend, an on-premises Python API server, and a local retrieval engine backed by curated JSONL corpora.

6.1 Frontend (`web/pampanito.html`)

A single-page application served directly by the backend. Designed for iOS Safari on iPhone as the primary visitor device, with full Android support.

- **Compartment-aware navigation** — 11 compartments + fore/aft deck, each with a unique ID passed to the API.
- **AudioContext playback** — GainNode normalises volume across narration, TTS answers, and pre-recorded fallback audio.
- **Voice input (dual path)** — On startup, the client calls `/health`; if `transcribe_available` is true, it routes all voice input through Groq Whisper (record → upload path). Otherwise it uses the browser's native Web Speech API. Browsers without any microphone support show a text input row.
- **Answer pipeline** — transcript POSTed to `/ask`; answer text POSTed to `/tts` for speech synthesis; audio streamed and played immediately.
- **No-cache headers** — visitors always load the latest tour version without a hard refresh.

6.2 API Server (`api/main.py`)

FastAPI application hosted with Uvicorn behind a self-signed TLS certificate (required for microphone access on mobile browsers).

Endpoint	Method	Purpose
`POST /ask`	JSON: `question_text`, `compartment_id`, `playhead_time_ms`	Main Q&A endpoint. Runs retrieval + extractive synthesis; returns a structured answer with citations and `partial_match` flag.
`POST /tts`	JSON: `text`, `language`	Converts answer text to speech; returns an MP3 audio stream.

Endpoint	Method	Purpose
`POST /transcribe`	Multipart: audio file	Android speech path — sends audio to Groq Whisper; returns transcribed text.
`POST /contact`	Multipart form	Captures visitor contact info and unanswered question for historian review.
`GET /health`	—	Returns server status including `transcribe_available` flag (whether a Groq API key is configured).
`GET /pampanito.html`	—	Serves the tour frontend.

6.3 Retrieval Engine

A custom token-overlap retrieval system with vocabulary expansion — no vector database or LLM required at query time. Designed for high accuracy on a closed, domain-specific corpus.

- **Tokenisation & stopword filtering** — query and corpus text are lowercased, punctuation-stripped, and filtered against a domain-tuned stopword list (including universal noise words like "submarine" that appear in every chunk, and prepositions such as "into", "upon", "through").
- **Synonym expansion (`QUERY_SYNONYMS`)** — query tokens are expanded with domain synonyms before scoring. For example, "eat" expands to *ate, galley, mess, food, meal*; "fired" expands to *launch, launched, tube*; "depth" expands to *deep, feet, dive, running, failure*. This bridges the vocabulary gap between visitor phrasing and corpus language.
- **Weighted corpus scoring** — Tour = 3.0×, FAQ = 1.2×, Shorts = 0.8×. Tour chunks in the visitor's current compartment score highest.
- **FAQ title-match bonus** — FAQ chunks whose question title covers all synonym-expanded query tokens receive up to a 4× coverage bonus (`max(weight, weight × 4.0 × coverage)`), ensuring the most precisely targeted FAQ answer wins over tangentially related tour passages.
- **Quantity boost** — for "how many" questions, chunks containing number words receive a 1.5× score multiplier.
- **Intent detection** — detects WHERE questions (prepends "In the [location]." to the answer), HOW MANY questions (activates quantity checking), and Mark 14/18 comparison questions (activates comparison-language boost only when both marks or explicit comparison vocabulary are present).

6.4 Extractive Answer Synthesis

Answers are assembled by extracting the most relevant sentences from the top-ranked corpus chunk — no LLM generation in the default deployment. This guarantees factual fidelity and eliminates hallucination risk.

- Sentences are filtered and ranked by synonym-expanded term overlap with the query.
- A second chunk is consulted if the first does not yield a complete answer.
- Speech filler words ("uh", "um", "er") are automatically stripped from oral-history transcript chunks before the answer is read aloud.
- A `partial_match` flag is returned when the answer is topically relevant but does not directly answer the question. The frontend plays a recorded "I don't have a direct answer, but here's what I know" prefix before reading the answer.

7. Answer Quality & Partial Match Logic

A key reliability feature is the system's ability to distinguish between a direct answer and a relevant-but-indirect answer.

Condition	Example
None of the core query terms appear in the assembled answer	Question about "escape hatches" but answer discusses hull construction
"How many X" question but no sentence contains both X and a number	"How many torpedoes" but answer only says they were stored forward
Question contains a superlative (worst, best, hardest…) but the answer does not address the judgment	"What was the worst bunk to sleep in" — answer describes bunks but cannot rank them

When `partial_match` is `true`, the frontend plays a soft spoken prefix before reading the answer, setting visitor expectations correctly. If no relevant content is found at all, a separate "nothing found" audio clip plays.

8. Deployment

Component	Detail
Hardware	Mac (MacBook Pro, Mac Mini, or equivalent). No GPU required.
Network	Local Wi-Fi access point. Visitors connect via QR code or direct URL. Internet not required except for optional Groq Whisper transcription.
TLS	Self-signed certificate on port 8443 — required by browsers for microphone access (HTTPS).
Python stack	Python 3.10+, FastAPI, Uvicorn, OpenAI SDK (TTS), Groq API (Whisper transcription). All dependencies in a local virtual environment.
Server startup	Single command: `bash start_https.sh` — kills any prior instance, sources environment variables, starts Uvicorn.
Environment	`.env.local` — `GROQ_API_KEY` for Android transcription, other service keys as needed.
LLM flag	`USE_LLM=false` (default) — fully local, no LLM calls for Q&A. API stub present for future GPT-based synthesis.

9. Current Status

The application is a fully functional prototype. It covers all 11 interior compartments plus the fore and aft decks of the Pampanito, and the AI guide can answer 226 custom-authored questions about the ship and her crew, plus hundreds more from the broader WWII submarine corpus.

Capability	Status
Voice Q&A — iOS / desktop	✓ Web Speech API
Voice Q&A — Android	✓ Groq Whisper record mode
Text input fallback	✓ All browsers
Multilingual TTS (6 languages)	✓
Compartment-aware retrieval	✓
Sub-2-second response time	✓
Honest partial-answer signalling	✓
226 custom Pampanito FAQs	✓ pam_001–pam_226
LLM-backed synthesis	Stubbed — not enabled

Near-term roadmap:

- Expand FAQ corpus to ~300 entries (batch 13+)
- Pre-cached TTS audio for the most common questions — instant playback without API latency
- Expanded corpus coverage: crew oral histories, patrol logs, post-war interviews
- QR code self-check-in for analytics (compartment traffic, popular questions)

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